

Rushin Bhatt

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EDUCATION

- Columbia University** New York, NY
 - Master of Science in Data Science (TA: COMS6998 Scalable Matrix Computation)* Aug 2025 – Dec 2026
 - Relevant Coursework:* Data Structures & Algorithms, Database, OS, Networks, Big Data, Cloud, Software Engineering, Probability, Machine Learning, Deep Neural Networks, Information Retrieval, Visualisation, NLP, GPU Computing, Agentic AI, AI Venture

SKILLS

Python (Pandas, NumPy, Matplotlib, Scikit-learn, TensorFlow, PyTorch, Selenium, FastAPI), C, R, JS, MERN, TypeScript, SQL, Git, AWS, GCP, Docker, AI (VLMs, RAG, Exa, CrewAI, Pydantic, LangGraph, LangChain, MCP), Vector Databases, CUDA, Cursor, Claude

EXPERIENCE

- Forward Deployed Engineering Intern – Spot AI** San Francisco, California, USA
 - Worked with customers on video agent deployment, balancing engineering, product, and sales.* May 2026 – Aug 2026
 - o **Agent Configuration:** Built evaluation pipelines and annotated datasets for A/B testing of query-to-video embedding models. Developed and shipped an internal agent iteration dashboard used by FDEs in production, boosting productivity by 3x.
 - o **Customer Engagement:** Constant calls with customers to understand problems, identify solutions, and deploy developments.
 - o **Product Feedback Loop:** Improved precision by over 45% on a customer use case by reducing FPs and parameter tuning.
- Machine Learning Intern – Fynd (Shopsense Retail Technologies Ltd.)** Mumbai, India
 - Handled multiple machine learning projects across two internship terms.* June – Aug 2024, Jan – May 2025
 - o **AI Catalog Enrichment:** Engineered an AI-driven catalog enrichment pipeline for brands like Veromoda & Tira, integrating VLMs & LLMs with search-based retrieval (Exa, SerpAPI, Perplexity), using LangGraph & CrewAI, which increased product attribute extraction accuracy by 37%. Worked directly under the CTPO, blending technical execution with product strategy.
 - o **Dynamic AOP:** Forecasting using XGB, AdaBoost, and ensemble models, boosting demand prediction precision by 20%.
 - o **Virtual Try-ons:** Implemented the SewFormer virtual try-on framework, enhancing garment fit prediction accuracy by 18%.
- Machine Learning Research & Development Intern – Dolcera Corporation** San Mateo, California, USA
 - Conducted research on maximizing the effectiveness of AI tools in development teams.* Dec 2024 – Jan 2025
 - o **LLM Productivity Optimization Research:** Conducted applied research on AI-assisted productivity tools, demonstrating a significant uplift in cross-team efficiency. Explored advanced prompt engineering techniques for LLMs by analyzing developer usage patterns across arXiv, HackerRank, Reddit, and GitHub, resulting in a 22% gain in response accuracy and consistency.
- Software Engineering Intern – Vistry** Ahmedabad, India
 - Built a chatbot for Vistry using OpenAI API with a Node.js backend.* Feb 2024 – April 2024
 - o **Chatbot Development:** Architected and launched a customer-facing chatbot using OpenAI APIs and Node.js, deployed on the company's production website and actively serving 2,000+ monthly users. Enhanced chatbot reliability by building automated error-handling mechanisms, achieving a 95% uptime during live usage. Linked the chatbot to WhatsApp using Twilio.
- Software Development Intern – Source Pro Infotech Pvt Ltd.** Ahmedabad, India
 - Developed a chatbot model for one of their clients using TensorFlow and NLTK in Python.* Oct 2023 – Feb 2024
 - o **Web Integration:** Constructed an enterprise-grade chatbot solution for client deployments, leveraging TensorFlow and NLTK in Python to deliver scalable conversational intelligence. Incorporated a web-based interface for the chatbot, boosting accessibility among client employees by 40%. Optimized accuracy by 30% through data refinement, including semantic preprocessing, stemming, and vocabulary integration.
- Data Analyst – SciKnowTech** Ahmedabad, India
 - Analyzed datasets including student information, fee transactions, and marksheets.* Aug 2022 – June 2023
 - o **Tools:** Processed large-scale student datasets, delivering insights that improved academic reporting accuracy by 25%. Automated data pipelines using Pandas, NumPy, Power BI, and Tableau, decreasing manual report preparation time by 40%.

PROJECTS

- **KickClone:** Built a semantic player retrieval system using cosine similarity over player embeddings in a vector database.
- **AutoRAGEvals:** A self-improving system automating RAG tuning across chunk size, retrieval strategy, top-k, and reranking.
- **AgentOpsBench:** Multi-agent framework for battery degradation, RUL, and sensitivity analysis using the NASA PCoE dataset.
- **MisfitAI:** An AI wardrobe MVP with weather and calendar-aware, LLM-driven outfit recommendations, onboarding 100 DAUs.
- **Tunecraft:** Analyzed and visualized song similarity using multi-layer deep neural networks and 3D tensor embedding projections.
- **FindMe:** An AI-agent powered platform that connects job seekers with recruiters, hiring managers, and potential referrers.
- **Zyber Chat:** Implemented MERN and developed a cipher-based locker and chat interface using the Kyber algorithm in Java.

RESEARCH (PUBLICATIONS)

- **Beyond Static Leaderboards: Predictive Validity for the Evaluation of LLM Agents:** Under Review - arXiv
- **Advancing the Computational Architecture of Digital Audio Workstations:** Under review – Computer Music Journal (MIT)
- **KickClone: A Model built to Revolutionize Football Scouting:** Published – IEEE
- **A Detailed Exploration of the Security Issues in Educational Applications:** Published – Springer Nature)
- **Enhancing Transportation Infrastructure: A Deep Learning Approach for Pothole Detection:** Published – IEEE
- **A Pioneering and Robust Technique for Optimising Aerial Route Planning:** Published – IJIMS (Inderscience)

ACHIEVEMENTS & INTERESTS

- Winner – Columbia StartMeUp Bootcamp; granted a funding of \$3000 to attend HumanX AI Conference in San Francisco.
- Member – Columbia Startup Lab (Downtown & Uptown), engaging with NYC's entrepreneurship and deep tech ecosystem.
- Attained distinction in all Trinity College of London Classical Piano, Music Theory, and Rock&Pop music examinations.
- Performed at 4 solo piano concerts, including a fundraiser that raised over \$6000 and a performance in front of Pope Francis.
- Represented Gujarat in the Reliance Cup soccer tournament at national level for 3 consecutive years.
- Selected as a Research Intern at IMT Brazil (São Paulo), to work on Autonomous Vehicles under Prof. Eduardo LL Cabral.
- Former Treasurer – Computer Society of India (PDEU Chapter), April 2023 to December 2023.